

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, Java, C++, SQL, Bash

Backend & Security

Node.js, Express, FastAPI, REST APIs, async services & queues, integration testing, authentication, rate limiting, audit logging, AWS KMS encryption, VAPT remediation

LLM & Agentic Systems

Agentic pipelines, LLM extraction & output verification, cell-level provenance, provider routing / fallback, evaluation & anti-Goodharting guardrails, RAG, OpenAI / Groq APIs, PyTorch

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Databases & Web

MongoDB, PostgreSQL, Redis, SQLite, React

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered a **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Owned backend security and data protection across the eTriage stack: hardened authentication and rate limiting, implemented **AWS KMS envelope encryption** for sensitive PII / clinical data, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings**.
- Drove reliability in production: monitored logs across releases and client integrations, diagnosed and resolved tenant-specific incidents, and added integration tests for regression protection in an Agile, fast-moving environment.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories and integrated LLM-powered context-aware replies into a chat product, improving type safety, maintainability, and end-user response quality.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that plans typed retrieval, reranks pages, classifies evidence regimes, extracts entities with deterministic parsers before LLM fallback, **verifies outputs**, and returns **cell-level provenance** for every result, tracing every extracted value back to its source.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction accuracy and ranking transparency.

AutoEval - Retrieval Evaluation Improvement Agent

github.com/Deva-1903/AutoEval

Python, IR evaluation tooling, Git automation

- Built an evaluation-improvement agent that proposes queries and labels, scores them against multiple retrieval systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective.
- Added **anti-Goodharting guardrails**, schema checks, duplicate rejection, and read-only holdout protection, and committed accepted updates back to Git with report cards for an auditable, reproducible evaluation workflow.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers, WebSockets, SQL

- Built an agent that fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time, prototyped end-to-end on Cloudflare Workers with embedded SQL storage.
- Designed durable workflow boundaries so multi-step tasks **recover gracefully** from process restarts and individual tool failures; agent state and conversation history persist server-side without an external database.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.